Panagiotis Aigaias Dimtsoudis

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Education

University of Pennsylvania — Philadelphia, PA
M.S.E. in Electrical Engineering (Submatriculation) — Expected May 2028
B.S.E. in Electrical Engineering | **GPA: 3.68/4.00** — Expected May 2028

Relevant Coursework: Digital Integrated Circuits and VLSI Fundamentals; Analog Integrated Circuits; Nanoscale Science and Engineering; Power Electronics; Fundamentals of Solid-State Circuits; Embedded Systems Laboratory.

Project Portfolio (click project titles for details)

SRAM-Based 4-bit MAC Accelerator — Custom 6T SRAM & Datapath

- Designed a 6T SRAM bitcell: β/γ sizing, DRC/LVS-clean layout, and post-Quantus hold/read SNM and write margin.
- Built 16×4 SRAM macros with 442 ps access: logical-effort row decoder, precharge, write drivers, and self-timed control.
- Built a three-stage MAC datapath: 4×4 multiplier, 12-bit accumulator, sequencer, and a custom 94 ps flip-flop.
- Closed timing at $T_{CLK,min} = 0.90$ ns, 0.8 V on extracted SRAM arrays: 505 μm^2, 0.39 pJ per MAC, 17 ns dot product.
- Automated SNM and write-margin extraction from Spectre exports in Python, cross-checked against a brute-force search.

Metal Detector PCB

- Designed and routed an analog PCB with ALD1103 MOSFETs, LC oscillators, a mixer, and filters.
- Compared bench measurements with LTspice: fixed oscillator at 50.2 kHz versus 50.27 kHz simulated.
- Assembled and soldered the PCB; used an oscilloscope to debug each stage and verify output to an 8 Ω speaker.

Wireless ECG Monitor — Analog Front-End & BLE

- Built an AD623 instrumentation-amplifier front end with virtual-ground biasing and high-pass/low-pass filtering.
- Measured approximately 61 dB maximum gain in stage testing and a 3 dB roll-off near 190.5 Hz.
- Streamed ECG over BLE at ∼200 Hz; wrote Python scripts for logging, plotting, and R-peak heart-rate estimation.

Analog Synthesizer — Two-VCO Monophonic Voice

- Built a monophonic synth voice: two VCOs, sub-osc, noise, mixer, resonant multimode VCF, two ADSRs, VCA, LFO.
- Designed every waveform from op-amps, not a synth IC: TL074 integrator, LM311 comparator, 2N3906 discharge, ±12 V.
- Produced saw, pulse/PWM, triangle, and sine with 1 V/oct tracking, octave/coarse/fine tuning, and hard sync.
- Measured ___% sine THD, a ___–___ Hz tuning range, ___ mV output noise, and ___ ppm/°C drift.
- Captured 11 Altium sheets with orderable parts, verified each in LTspice, then laid out and assembled the PCB.

Guitar Hero Synthesizer

- Wrote bare-metal C for an ATmega328PB synthesizer with PWM audio, ADC whammy input, and an I2C LCD.
- Resolved speaker-induced shutdowns and audio noise using a 12 V battery and two 5 V regulators.

Experience

Device Research and Engineering Laboratory — Philadelphia, PA
Research Assistant — May 2026 – Present

- Characterized 6 nm AlN/AlScN/AlN junctions from 14–400 K using J–V, C–V, PUND, breakdown, and endurance tests.
- Developed Python scripts to extract device parameters and compare switching, retention, and endurance.
- Contributed measurements and analysis to a manuscript in preparation on cryogenic all-nitride ferroelectric memory.
- Built 3D CAD models of the device stack and an 8×8 crossbar array to support compute-in-memory design.

ESE 2180: Electronic, Photonic, and Electromechanical Devices — Philadelphia, PA
Teaching Assistant — Aug 2026 – Present

- Develop homework assignments, lab worksheets, and exams for a device physics course of 80+ students.
- Set up equipment and teach labs, connecting device physics to hands-on experiments.
- Hold weekly office hours on transistors, LEDs, and photovoltaics; grade work and explain concepts through feedback.

Automation Systems Hellas — Thessaloniki, Greece
Computer Engineering Intern — Jun 2024 – Aug 2024

- Developed C and Python utilities for log parsing and configuration checks to support commissioning.
- Updated wiring and I/O documentation, imaged PCs and HMIs, and monitored servers and network devices.

Skills

Programming: C/C++, Python, MATLAB, Java, Assembly, OCaml
Embedded Systems: ATmega328PB, UART, I2C, ADC, PWM, BLE
Hardware: Analog circuit design, PCB layout, soldering, circuit debugging
Test & Characterization: Oscilloscopes, cryogenic probe stations, J–V, C–V, PUND, endurance testing
Design Tools: Cadence Virtuoso, Spectre, Pegasus DRC/LVS, Quantus, Altium Designer, LTspice, SolidWorks, Fusion 360
Languages: Greek (Native), English (Fluent), German (Intermediate), Japanese (Intermediate)